Covalent Organic Frameworks for Photocatalytic Hydrogen Evolution:
A Factor-Chain Survey and a Synthesis-Design Bridge to TpPa-1

GoAI Research Multi-Agent Survey System

(competition run, 2026-08-27; all citations machine-audited)

Abstract

Covalent organic frameworks (COFs) have progressed from the first photocatalytic hydrogen-evolution demonstration in 2014 to apparent quantum efficiencies above 80% at 450 nm, yet the literature remains organized by material family rather than by the design levers that actually move performance. We survey 37 machine-audited primary studies and reviews through a factor-chain taxonomy: linkage chemistry (L1) sets the stability–conjugation envelope; building-block electronics (L2) set light harvesting; exciton management and crystallinity (L3) determine carrier delivery; the catalytic interface (L4) converts electrons to H_2; and synthesis methodology (L5) determines which materials exist at what quality. On this evidence base we register six research gaps with per-claim citation chains and database provenance, and we convert the most actionable gap — the missing link between green, scalable synthesis routes and photocatalytic performance — into a decision-recorded synthesis plan for the β-ketoenamine COF TpPa-1: a solvothermal baseline with two quantified process optimizations (microwave: 72 h→1 h with a ∼4.8× surface-area gain; green-solvent flow: 30× space-time yield at 89% lower specific energy), a thermodynamic feasibility audit with an explicit to-be-computed list, and a mandatory safety annex. The survey demonstrates a reusable template for turning literature evidence into auditable synthesis decisions.

1 Introduction

Covalent organic frameworks (COFs) entered materials chemistry in 2005, when boronate-ester condensation delivered the first crystalline, porous, fully organic frameworks with pore sizes of 7–27 Å and high thermal stability [6]. Photocatalysis arrived nine years later: a hydrazone-linked TFPT-COF bearing 3.8 nm mesopores produced hydrogen from water under visible light with a Pt cocatalyst, establishing COFs as a designable photocatalyst class [27]. The decade since has been one of rapid quantitative progress: linkage-protonated donor–acceptor (D–A) COFs now reach hydrogen evolution rates of 20.7 $\mathrm{mmol\,g^{-1}\,h^{-1}}$ [34], D–A COF nanosheets report apparent quantum efficiencies (AQE) of 82.6% at 450 nm [16], and framework reconstruction strategies push sacrificial rates to 27.98 $\mathrm{mmol\,h^{-1}\,g^{-1}}$ [35].

Existing reviews organize this literature primarily by material family or application class [4, 29]. That organization answers "which COF has been made," but not the question a synthetic chemist actually faces: *which design lever moved the number, and what should I therefore synthesize next?* Mechanistic perspectives have begun to frame COF photocatalysis as a pipeline — photon absorption, exciton dissociation, charge transport, and interfacial reduction — whose weakest stage caps the overall rate [3]. This survey adopts that pipeline view and contributes:

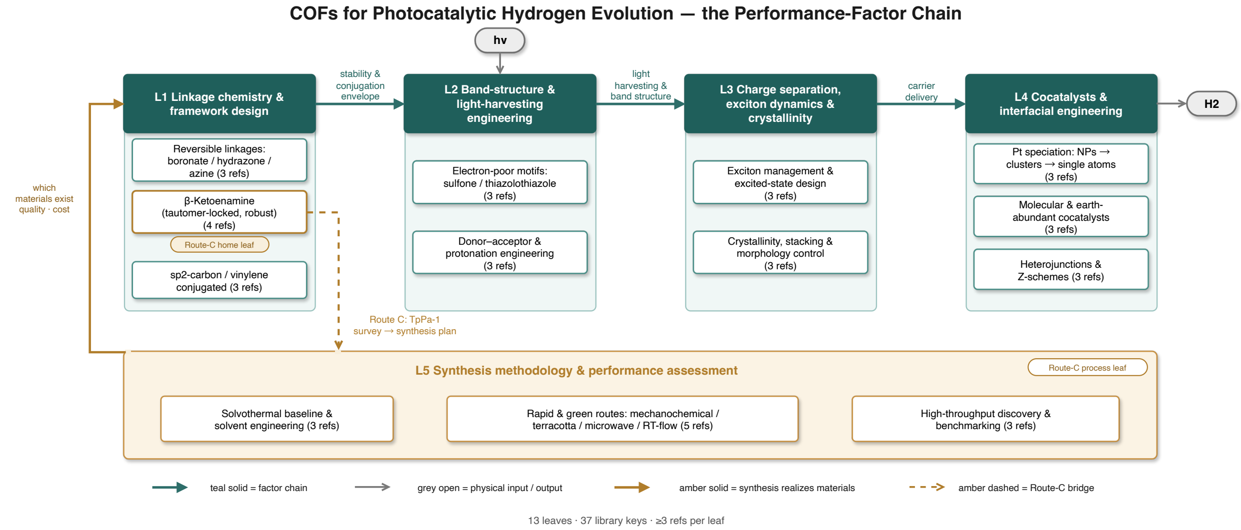

Figure 1: Factor-chain taxonomy of COF photocatalysts for the hydrogen evolution reaction (HER). Top: five groups of design levers (L1–L5) identified in this survey; bottom: the photon-to-H_2 physical pipeline. Labeled arrows mark the pipeline stage each lever family principally governs; the synthesis level (L5) acts indirectly by setting crystallinity and phase purity.

- **C1 — A factor-chain taxonomy** (Fig. 1). Five lever groups — linkage chemistry (L1), band-structure engineering (L2), exciton and crystallinity management (L3), interfacial catalysis (L4), and synthesis methodology (L5) — are mapped onto the pipeline stage each principally governs, with at least three primary references per taxonomy leaf (Secs. 2–4).
- **C2 — An evidence-chained gap register.** Six research gaps, each grounded in at least two audited library references with database provenance and explicit reasoning about why current work leaves the gap open (Sec. 6).
- **C3 — A survey-to-synthesis bridge.** A decision-recorded synthesis and process-optimization plan for the β-ketoenamine COF TpPa-1, converting the factor-chain evidence into an executable, safety-annotated protocol with a quantified route comparison and a thermodynamic feasibility audit (Sec. 5).

Scope follows the audited protocol of this project: crystalline 2D/3D COFs evaluated for sacrificial photocatalytic HER, plus synthesis-methodology studies of the TpPa family required by C3; metal–organic frameworks, CO_2 photoreduction, and pure oxygen-evolution studies are excluded. All 37 cited references were retrieved by multi-source search (arXiv, OpenAlex, Crossref) and passed a zero-tolerance citation-integrity audit; the audit trail is part of the deliverable. The survey is written for materials and chemistry graduate students: the intended takeaway is a mental model of *which factor limits which performance regime*, plus a reusable template for turning literature evidence into a synthesis decision.

2 Design levers I: linkage chemistry and band-structure engineering

2.1 L1 — Linkage chemistry sets the stability–crystallinity envelope

The linkage is the first and most consequential design decision, because it fixes both the chemical stability envelope and the degree of π-conjugation available to every downstream lever. COF linkage design is governed by a now well-articulated trade-off: reversible bond formation enables the error correction that produces crystallinity, but the same reversibility makes the framework hydrolytically fragile [12]. The field's linkage history is a sequence of answers to this trilemma.

Reversible foundations (L1a). Boronate-ester frameworks demonstrated that purely organic building blocks can crystallize into predictable lattices [6], but their boron–oxygen bonds hydrolyze readily, which confines them to anhydrous applications [6, 12]. Hydrazone linkages brought the first photocatalytic milestone: TFPT-COF evolved hydrogen from water with a sacrificial donor and Pt cocatalyst [27]. Azine linkages then delivered an early systematic structure–activity series: in the N_x-COF platform, HER activity rises monotonically with the number of ring nitrogens (N3 > N2 > N1 > N0), demonstrating that atomically precise framework chemistry translates into tunable photocatalysis [28].

β-Ketoenamine lock (L1b). The TpPa family resolved the stability half of the trilemma with a two-stage thermodynamic design: reversible Schiff-base condensation of 1,3,5-triformylphloroglucinol (Tp) with *p*-phenylenediamine (Pa-1) in 1:1 mesitylene/dioxane first builds a crystalline enol-imine network, which then undergoes an *irreversible* enol→keto tautomerization; only the keto form is observed, and the resulting TpPa-1 survives 9 N HCl and boiling water [14]. Because the lock is chemical rather than processual, it survives radically different syntheses: room-temperature solvent-free grinding gives the same stable frameworks, albeit with moderate crystallinity and an exfoliated, graphene-like morphology [2]. The scaffold is also a productive photocatalysis platform: acetylene- and diacetylene-walled β-ketoenamine COFs show that extending wall conjugation markedly raises HER (the diacetylene analogue outperforms the acetylene one) [22], and a fixed-backbone substituent series (TpPa-COF-X, X = –H, $-(CH_3)_2$, $-NO_2$) maps how peripheral electronics shift morphology, light absorption, and activity [26]. This chemistry is the home leaf of the Route C target (Sec. 5).

Fully conjugated linkages (L1c). Vinylene (sp^2-carbon) linkages remove the conjugation bottleneck of C–N single bonds. sp^2-carbon-conjugated frameworks have been applied to photocatalytic hydrogen production [13]; mild Knoevenagel polycondensation now affords vinylene-linked benzobisthiazole frameworks (BTH-1/2/3) [31], and benzothiazole-mediated aldol-type polycondensation yields unsubstituted-vinylene 2D COFs whose two-dimensional frameworks outperform their 1D counterparts in photocatalysis [17]. The cost is narrower monomer scope and condition-sensitive synthesis [17, 31].

Read as a series, L1 shows the trilemma being solved twice over: the β-ketoenamine route buys hydrolytic stability by spending linkage reversibility *after* crystallization has already happened [14], whereas the vinylene route accepts a harder crystallization problem up front in exchange for uninterrupted conjugation [12, 17]. Which trade is better is not decidable at the linkage level alone — it depends on whether the downstream bottleneck is carrier transport or framework survival, which is precisely why this survey tracks the full factor chain rather than ranking linkages in isolation [12, 35].

2.2 L2 — Band-structure and light-harvesting engineering

With the linkage fixed, the second lever tunes what the framework absorbs and how strongly photogenerated excitons are pulled apart. Two motifs dominate the library evidence.

Electron-poor heteroaromatics (L2a). Sulfone chemistry produced one of the field's benchmark materials: the dibenzothiophene-sulfone framework FS-COF for hydrogen evolution from water [30]. Its significance was consolidated by a factor-deconvolution study on the same family, which varied band gap, crystallinity, surface area, exciton migration, and charge extraction quasi-independently and identified crystallinity and charge extraction as the prime factors —

Table 1: Linkage classes for COF HER photocatalysts (taxonomy level L1). Stability and activity statements are those reported in the cited studies.

Linkage class	Exemplar	Formation chemistry	Reported stability / HER relevance
Boronate ester (L1a)	COF-1, COF-5 [6]	Boronic acid + catechol condensation	Thermally robust; hydrolytically labile [6, 12]
Hydrazone (L1a)	TFPT-COF [27]	Aldehyde + hydrazide	First COF HER demonstration [27]
Azine (L1a)	N_x-COF [28]	Aldehyde + hydrazine	Water/photo-stable; HER scales with N count [28]
β-Ketoenamine (L1b)	TpPa-1/-2 [14]	Reversible imine + irreversible tautomerization	9 N HCl, boiling water (9 N NaOH for TpPa-2) [14]; HER platform [22, 26]
Vinylene / sp^2-C (L1c)	BTH-1/2/3 [31]; v-2D-COF-NS [17]	Knoevenagel / aldol-type polycondensation	Full in-plane conjugation; 2D > 1D photocatalysis [17]

a rare causal, rather than correlational, result in this literature [10]. Thiazolo[5,4-d]thiazole acceptors paired with pyrene donors (PyTz-COF) show that a bicontinuous donor/acceptor channel architecture underlies the exceptional optoelectronic response [18].

Donor–acceptor construction and post-synthetic protonation (L2b). Multicomponent synthesis extends the D–A idea to three components: triazine acceptors and benzotrithiophene donors bridged by sp^2-carbon links boost HER beyond two-component controls [21]. Post-synthetic linkage protonation of imine D–A COFs red-shifts absorption and improves charge separation, reaching $20.7\ \mathrm{mmol\,g^{-1}\,h^{-1}}$ [34]. In exfoliated form, a β-ketene–cyano D–A pair in COF nanosheets couples dramatically prolonged carrier lifetimes (femtosecond transient absorption evidence) with an AQE of 82.6% at 450 nm — currently the clearest demonstration that L2 engineering, executed on an L1-stable scaffold and read out through L3 photophysics, produces step-change activity [16].

3 Design levers II: charge management and interfacial catalysis

3.1 L3 — Excitons, crystallinity, and stacking order

Absorbed photons in organic frameworks yield strongly bound excitons, so the third lever is whether those excitons dissociate and whether the resulting charges survive transport. Mechanistic perspectives now treat photon absorption, exciton dissociation, and electron delivery as separately measurable stages of one pipeline [3].

Excited-state design (L3a). Computation has moved upstream of synthesis: density-functional screening across donor–acceptor pairs shows the exciton binding energy E_b is systematically tunable through D–A strength, making E_b a design variable rather than a post-hoc measurement [24]. On the experimental side, an isoreticular hydrazide-COF series resolved how excited-state electron distribution and transport pathways change with the linker, tying photophysics to specific framework positions [5]. What is still missing is the closed loop between the two communities — computed E_b values validated by spectroscopy on the same material series (Sec. 6, G5) [3, 5, 24].

Crystallinity and stacking (L3b). The factor-deconvolution benchmark identified crystallinity as a prime HER factor [10], and two complementary studies show why and how to act on it. During photocatalytic cycling, coaxial stacking can disorder and activity decays; threading poly(ethylene glycol) through the mesopores locks the stacked configuration and prevents the decline [37] —

direct evidence that *operando* structural integrity, not just as-made crystallinity, matters. At the synthesis end, the reconstruction strategy decouples polymerization from crystallization: amorphous imine networks pre-organized by a removable covalent tether reconstruct into highly crystalline frameworks, lifting sacrificial HER to 27.98 $\mathrm{mmol\,h^{-1}\,g^{-1}}$ — the highest sacrificial rate in this survey's library — while confirming that poor reversibility of robust linkages is the central obstacle the field must engineer around [12, 35].

3.2 L4 — Cocatalysts and interfacial engineering

Even perfectly delivered electrons produce no hydrogen without an interface that turns them over. The library evidence organizes into three interface strategies.

Pt speciation (L4a). Platinum remains the default electron sink, but its speciation is now a lever in its own right: dispersing Pt as single atoms on a COF raises active-site density relative to nanoparticles [8], while in-situ photodeposition yields cluster-scale Pt with strong electronic coupling to the framework [19]. The factor-deconvolution protocol treats Pt loading and speciation as one of the deconvoluted variables, which is what makes cross-material activity comparisons meaningful at all [10].

Molecular and earth-abundant cocatalysts (L4b). The azine N_2-COF paired with a chloro(pyridine)cobaloxime delivers 782 $\mu\mathrm{mol\,h^{-1}\,g^{-1}}$ at a turnover number of 54.4, with mechanistic analysis supporting outer-sphere electron transfer to a monometallic cobalt pathway [1]. Covalently wiring the cobaloxime to the backbone improves and prolongs activity relative to physisorbed analogues by enabling recoordination of the cocatalyst during operation [11]. These molecular systems remain orders of magnitude below Pt-based benchmarks in rate and turnover (cf. [34]), a gap analyzed in Sec. 6 (G6) [1, 11]. Interface design more broadly — including cocatalyst anchoring chemistry — is a central theme of the 2024 Accounts survey [4].

The sacrificial-system context. Almost all quantitative claims in this section are measured against sacrificial electron donors, and the donor is itself an interface-level variable: the cobaloxime/N_2-COF benchmark operates with triethanolamine in a water/acetonitrile medium [1], while the record-setting protonated D–A systems were developed in ascorbic-acid media, where the donor also participates in the protonation equilibrium that red-shifts absorption [34]. Cross-study comparisons that ignore the donor identity therefore conflate interfacial kinetics with donor thermodynamics — one of the reasons a fixed-protocol benchmark is registered as gap G1 in Sec. 6 [1, 10].

Heterojunctions and Z-schemes (L4c). Coupling COFs to inorganic partners re-routes charges across a built-in junction: an in-situ grown, chemically bonded 2D/2D COF / oxygen-vacancy-WO_3 Z-scheme targets *overall* water splitting rather than sacrificial HER [25], and a 2D/2D COF/CdS Z-scheme heterojunction enhances photocatalytic hydrogen evolution [9]. Composite and heterojunction photocatalysts are surveyed systematically in the 2020 CSR review [29]. The overall-water-splitting rates remain far below sacrificial benchmarks [25], marking the sacrificial-donor dependency as a field-level open problem (Sec. 6, G2).

Taken together, L3 and L4 close the causal chain begun in Sec. 2: linkage and band engineering determine what can be absorbed and separated; crystallinity, stacking, and the catalytic interface determine how much of that potential is converted. The prime-factor evidence [10] justifies the ordering used in the synthesis design of Sec. 5: crystallinity first, interface second, everything else after.

4 Synthesis methodology: five routes to the same framework

The final lever (L5) is often treated as logistics, but the library evidence says otherwise: the synthesis route sets the crystallinity and phase purity on which every upstream factor depends [10, 35]. This section compares the five route families documented for the TpPa system, the target chemistry of Route C.

Solvothermal baseline and solvent thermodynamics (L5a). The canonical TpPa-1 protocol — Tp + Pa-1 in 1:1 mesitylene/dioxane with aqueous acetic acid, sealed at 120 °C for 72 h — remains the reference for crystallinity and chemical robustness [14]. Solvent choice, long empirical, now has a quantitative decision framework: ranking solvents by CHEM21 greenness, Hansen solubility parameters, and free energy of solvation shows that moderate-solvation media (1,4-dioxane, propylene carbonate, diacetin) suppress excessive reaction reversibility and promote crystallite growth [33]. New synthetic strategies across the COF field are compiled in the dedicated CSR review [20].

Rapid and green routes (L5b). Four alternatives trade conditions against quality:

- *Mechanochemical grinding*: room-temperature, solvent-free, delivering the same chemical stability with moderate crystallinity and partially exfoliated morphology [2].
- *Salt-mediated ("organic terracotta")*: p-toluenesulfonic acid mediation crystallizes ultraporous frameworks in seconds, with family surface areas up to 3000 $m^2\,g^{-1}$ and continuous processing demonstrated on a twin-screw extruder [15].
- *Microwave-assisted solvothermal*: an enamine COF forms in significantly less time at lower temperature with high yield [32]; in the direct TpPa-1 comparison compiled by the microwave review, 1 h at 100 °C gives a BET surface area of 725 $m^2\,g^{-1}$ versus 152 $m^2\,g^{-1}$ for the conventional 3-day, 120 °C control [7]. The same review documents an instructive caveat: for transimination-based crystallinity upgrading, solvothermal heating still outperforms microwave for β-ketoenamine COFs — the "fastest route" is context-dependent [7].
- *Room-temperature and flow synthesis*: RT batch TpPa-1 matches solvothermal crystallinity and porosity, attributed to monomer solubility and π-stacking-assisted error correction; the first continuous-flow COF production reached 41 $mg\,h^{-1}$ at a space-time yield of 703 $kg\,m^{-3}\,day^{-1}$ [23]. Flow synthesis of TpPa-1 in the green solvent diacetin reaches a BET area of 418 $m^2\,g^{-1}$ with a 30-fold space-time-yield increase and an 89% reduction in specific energy relative to batch in the same solvent [33].

High-throughput discovery (L5c). Scale-out of synthesis itself is emerging: sonochemical high-throughput experimentation screened 76 polymers, of which 60 formed crystalline COFs and 18 were previously unreported, in a single campaign — the target application was photocatalytic H_2O_2 production, but the accelerated synthesis-plus-screening methodology transfers directly to HER [36]. The campaign format matters for this survey's argument: it is the first library-scale demonstration that route parameters (here, sonication) can be co-screened with photocatalytic output, rather than fixed by convention before testing begins [36]. Combined with AQE-standardized reporting [16] and factor benchmarking [10], this points toward synthesis–performance maps built at campaign scale rather than paper scale.

The pattern across Table 2 motivates the design task of the next section: no single route dominates all axes, and the HER consequences of the green/scalable routes are essentially unmeasured (Sec. 6, G4). Choosing a route is therefore a decision problem with documented trade-offs — exactly the kind of problem a survey should hand over to synthesis planning in executable form.

Table 2: Synthesis routes to TpPa-family COFs (taxonomy level L5). Quantities are as reported in the cited sources; dashes mark values not reported in the library evidence.

Route	Conditions	Reported quality	Scale/greenness notes
Solvothermal [14]	120 °C, 72 h, mesitylene/dioxane + aq. AcOH	Reference crystallinity; 9 N HCl and boiling-water stability	Sealed-tube batch; high-boiling solvents
Mechanochemical [2]	Room temperature, solvent-free grinding	Moderate crystallinity; stability retained; exfoliated layers	Greenest inputs; lowest crystallinity
Salt-mediated [15]	p-TsOH mediation, seconds	Highly crystalline, ultraporous (up to 3000 $m^2\,g^{-1}$, family value)	Twin-screw extrusion; salt work-up required
Microwave [7, 32]	100 °C, 1 h	BET 725 vs. 152 $m^2\,g^{-1}$ (conventional control)	Fastest lab route; penetration-depth scale limit
RT / flow [23, 33]	RT batch; flow in diacetin	RT: comparable to solvothermal; flow: BET 418 $m^2\,g^{-1}$	Flow: 30× STY, −89% specific energy; COF flow precedent 703 $kg\,m^{-3}\,day^{-1}$

5 Synthesis design (Route C): from factor chain to a TpPa-1 protocol

This section converts the survey's evidence into a decision-recorded synthesis plan for the β-ketoenamine COF TpPa-1 (monomers: 1,3,5-triformylphloroglucinol, Tp; and *p*-phenylenediamine, Pa-1; stoichiometry 2 Tp : 3 Pa-1 from the 3×CHO / 2×NH_2 functional balance) [14]. The full safety-annotated protocol, candidate-route decision table, and machine-readable experiment plan are deliverable artifacts of this project;[1] this section presents the decision logic and the quantitative core. Figure 2 shows the route.

Why TpPa-1, and why these criteria. The factor chain orders the design requirements: crystallinity and charge extraction are the prime activity factors [10], and aqueous photocatalysis presupposes hydrolytic stability, which the irreversible enol→keto lock provides at the linkage level [12, 14]. TpPa-1 is also the only COF in the library with five independently documented synthesis routes (Table 2), making it the strongest available testbed for evidence-based process selection [2, 14, 15, 23, 32, 33].

Baseline protocol (decision: solvothermal R1). Monomer quality first: Pa-1 is purified (sublimation) immediately before use — a chemical-practice inference consistent with the observation that monomer solubility and purity enable mild-condition growth [23]; Tp is purchased or prepared by the literature Duff-type formylation of phloroglucinol [14]. Condensation follows the origin protocol (1:1 mesitylene/dioxane, aqueous AcOH, 120 °C, 72 h, sealed tube) [14], chosen over the four alternatives because it carries the richest characterization record against which optimized batches can be compared (Table 2). Work-up uses solvent washing, solvent exchange, and vacuum activation [14, 15]. Deployment for HER follows the TpPa-COF-X benchmark configuration (in-situ Pt photodeposition, sacrificial donor) [26], with Pt speciation tunable as an independent factor [8, 19].

[1] Artifacts: `ideas/route_c_synthesis_plan.md` and `ideas/experiment_route_c_tppa1.json` in the project workspace.

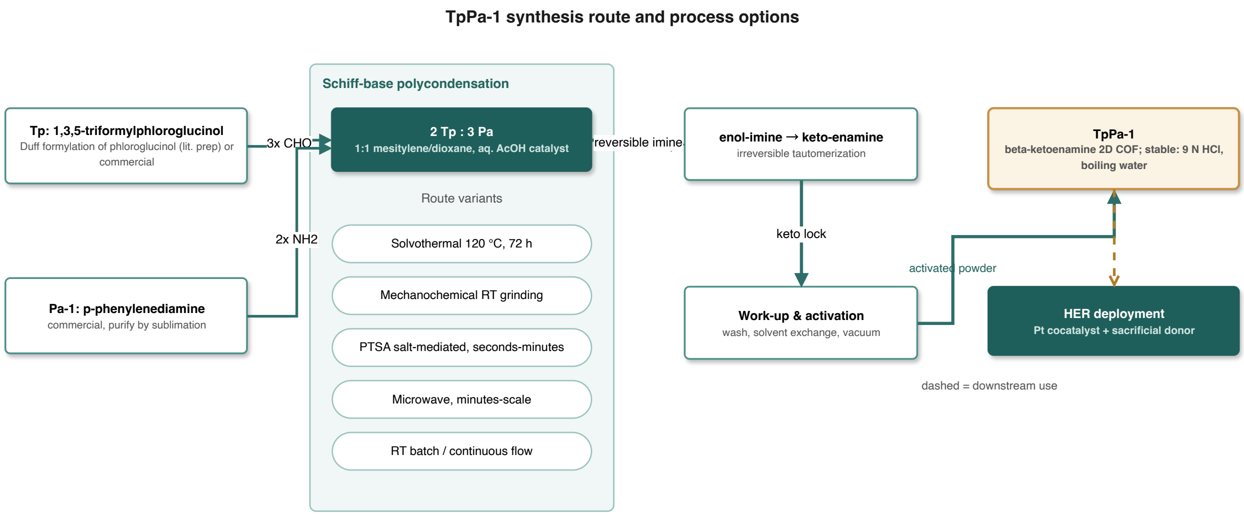

Figure 2: Synthesis of TpPa-1. Monomers condense via reversible Schiff-base chemistry (route variants shown as chips; quantitative comparison in Table 2) and lock into the β-ketoenamine form by irreversible tautomerization, giving an acid/base/boiling-water-stable framework deployable for photocatalytic HER.

Optimization O1 — microwave heating. Replacing oven heating with microwave irradiation reduces reaction time from 72 h to 1 h and temperature from 120 to 100 °C while *increasing* the BET surface area from 152 to 725 $m^2\,g^{-1}$ in the direct comparison [7, 32] — a 98.6% time reduction and a $\sim$4.8$\times$ porosity gain. The documented caveat (solvothermal beats microwave for transimination-based upgrading of β-ketoenamine COFs [7]) is recorded in the decision table: O1 optimizes the *direct* route, not every quality pathway.

Optimization O2 — green-solvent continuous flow. The solvent-thermodynamics framework (CHEM21 greenness, Hansen parameters, free energy of solvation) selects diacetin; flow-derived TpPa-1 reaches a BET area of 418 $m^2\,g^{-1}$, a 30-fold space-time-yield increase, and an 89% reduction in specific energy versus batch in the same solvent [33]. Feasibility rests on the demonstrated room-temperature growth of TpPa-1 with solvothermal-grade quality and on the flow precedent of 703 $kg\,m^{-3}\,day^{-1}$ [23]. A salt-mediated seconds-scale route is held in reserve for crystallinity rescue at scale [15], with amorphous-to-crystalline reconstruction as a further fallback paradigm [35].

Thermodynamic feasibility audit. Literature-supported statements: (i) the route is a reversible imine equilibrium (error-correcting, crystallinity-enabling) feeding an irreversible tautomerization whose keto product is the only observed form — a thermodynamic sink [14]; (ii) retained crystallinity in 9 N HCl and boiling water evidences the depth of that sink [2, 14]; (iii) moderate-solvation media thermodynamically balance reversibility against crystallite growth [33]; (iv) the general reversibility–crystallinity–stability trade-off frames why this two-stage design is necessary at all [12, 35]. Quantities *not* reported in the library — keto/enol tautomer energetics, per-bond condensation free energy, periodic formation and stacking energies, monomer solvation free energies — are deliberately left as a to-be-computed list with recommended DFT setups (dispersion-corrected hybrid functionals with implicit solvation for molecular models; dispersion-corrected periodic PBE for the framework; COSMO-RS for solvation screening) rather than asserted numerically. No numeric value in this section originates outside the cited sources.

Validation and characterization plan. Every batch is judged against literature-anchored acceptance criteria: powder X-ray diffraction against the simulated pattern establishes crystallinity [14]; FT-IR and ^{13}C CP-MAS NMR must show the keto-form signature that certifies complete tautomerization [14]; nitrogen sorption benchmarks porosity against the 725 $m^2 g^{-1}$ (microwave) and 418 $m^2 g^{-1}$ (flow) reference values [7, 33]; and a 9 N HCl / boiling-water soak with post-treatment diffraction reproduces the origin stability test [14]. Stacking disorder during photocatalytic cycling is monitored as a separate failure mode, following the operando evidence [37]. Because the three production routes (R1, O1, O2) will be characterized under one protocol and then deployed under the same HER configuration [26], the plan doubles as the seed experiment for gap G4 — the missing route→crystallinity→HER chain (Sec. 6) [10].

Safety and integration notes. The deliverable protocol carries a mandatory hazard table (diamine sensitizer, dioxane peroxide former, sealed-vessel pressure, microwave-rated reactors only [7], H_2/O_2 explosion limits during photocatalysis, acid handling for in-house Tp preparation). A retrosynthesis-backend integration demo (stub provider) is included in the project state *explicitly labeled as demonstration data, not chemical conclusions*; all chemistry above traces to the audited literature.

6 Open challenges: an evidence-chained gap register

Each gap below is grounded in at least two audited references, annotated with the database that surfaced the evidence (arXiv / OpenAlex / Crossref, per the project search log), and closed with the reasoning why current work leaves it open. The register is maintained as a standalone deliverable (`notes/research_gaps.md`) and summarized here.

G1 — No standardized cross-family HER benchmark. Factor deconvolution exists within a single family (S-COF/FS-COF; OpenAlex) [10], while performance is reported in incommensurable currencies — AQE at a stated wavelength (OpenAlex) [16] versus mass-normalized rates (OpenAlex) [34] — and high-throughput screening has targeted H_2O_2 rather than HER (OpenAlex) [36]. *Why open:* multi-variable control samples are costly, and reporting conventions differ across groups — the same library contains AQE-denominated and mass-rate-denominated headline results with no conversion basis [16, 34]; no library study runs two linkage families under one fixed protocol. *Needed:* a fixed-protocol, multi-family benchmark reporting both AQE and mass rate.

G2 — Overall water splitting remains a single-example regime. The only library system targeting non-sacrificial operation is the chemically bonded 2D/2D COF/WO_3 Z-scheme, with rates far below sacrificial benchmarks (OpenAlex) [25]; the 2020 CSR review confirms sacrificial dominance across COF photocatalysis (OpenAlex/Crossref) [29]; even molecular-cocatalyst systems rely on sacrificial donors (OpenAlex) [1]. *Why open:* sacrificial donors mask the four-electron oxidation half-reaction, whose oxidative stress and band-position demands most COFs were never designed to meet. *Needed:* valence-band and anti-oxidation engineering plus Z-scheme charge routing designed for the OER half-reaction.

G3 — Operando structural durability is unquantified. Coaxial stacking disorders during photocatalytic cycling and activity decays unless the stack is locked (OpenAlex) [37]; chemical stability tests (9 N HCl, boiling water; Crossref) [14] probe a different failure axis, and the linkage-design trilemma contains no operando dimension (OpenAlex) [12]. *Why open:* the field equates chemical with photocatalytic stability, and standard practice is before/after PXRD rather than time-resolved structural tracking. *Needed:* operando structure–activity correlation (cycles vs. crystallinity vs. rate) and a shared “photocatalytic lifetime” reporting convention.

G4 — Green and scalable routes are not HER-benchmarked. Flow-derived TpPa-1 reports BET area, space-time yield, and energy metrics but no HER data (Crossref) [33]; the microwave comparison stops at porosity and CO_2 uptake (OpenAlex) [7, 32]; room-temperature and flow routes close with crystallinity and porosity (OpenAlex) [23]. *Why open:* the synthesis methodology and photocatalysis communities publish to different completion criteria, so the route→crystallinity→HER chain — whose middle link is causally established (OpenAlex) [10] — has never been closed on one material. *Needed:* same-protocol HER comparison of solvothermal, microwave, flow, and mechanochemical TpPa-1 — precisely the experiment Route C (Sec. 5) is designed to seed.

G5 — Exciton-binding-energy design rules lack an experimental loop. DFT screening makes E_b tunable on paper (OpenAlex) [24]; excited-state mapping is family-specific (OpenAlex) [5]; and the mechanistic perspective explicitly calls the pipeline's characterization methodology unsettled (OpenAlex) [3]. *Why open:* measuring E_b on polycrystalline powders is noisy and mismatched to idealized computational models, and computational and spectroscopic groups rarely share sample sets. *Needed:* a computed-then-measured E_b benchmark across one isoreticular series.

G6 — Earth-abundant cocatalysts trail Pt by orders of magnitude. The cobaloxime/N_2-COF system achieves 782 $\mu\mathrm{mol\,h^{-1}\,g^{-1}}$ at TON 54.4 (OpenAlex) [1]; covalent wiring prolongs but does not eliminate degradation (OpenAlex) [11]; meanwhile the field's effort concentrates on Pt speciation (OpenAlex) [8, 19]. *Why open:* molecular-cocatalyst failure modes demand operando spectro-electrochemistry beyond most COF groups' stacks, while Pt photodeposition is operationally trivial. *Needed:* self-healing ligand design on covalent anchors, and lifecycle TON accounting that makes the Pt/non-Pt comparison honest.

These six gaps are not independent: G1's protocol deficit blocks progress measurement on G2–G6, and G4 is the gap this survey's Route C plan directly operationalizes. The gap register with full evidence chains and database provenance accompanies this survey as an audited deliverable.

7 Conclusion

This survey organized COF photocatalytic hydrogen evolution by performance factor rather than material family. The factor chain — linkage chemistry (L1), band engineering (L2), exciton and order management (L3), interfacial catalysis (L4), synthesis methodology (L5) — maps each design lever to the pipeline stage it governs, with crystallinity and charge extraction established as the prime factors by the field's clearest causal benchmark [10]. Along that chain, the β-ketoenamine tautomer lock stands out as the load-bearing chemical innovation for aqueous operation: a reversible, error-correcting condensation that ends in an irreversible, hydrolysis-proof sink [14].

The survey's second half converted that evidence into practice. The six-entry gap register shows where the literature stops — most actionably, the green and scalable synthesis routes whose HER consequences remain unmeasured — and the Route C plan turns that specific gap into an executable protocol: a solvothermal baseline with two quantified optimizations (microwave: 1 h at 100 °C with a ∼4.8× porosity gain; green-solvent flow: 30× space-time yield at −89% specific energy), a thermodynamic audit that refuses to invent numbers, and a mandatory safety annex [7, 33].

The reader we designed for should leave with two assets: a mental model of which factor limits which performance regime, and a worked template for converting survey evidence into a synthesis decision — the survey-to-synthesis bridge that the COF hydrogen-evolution field, now rich in materials and mechanisms but poor in standardized decisions, most needs.

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